PREDICTING FOOTBALL PLAYER MARKET VALUE

Position-Specific Machine Learning Pipeline

DAMA Hackathon 26

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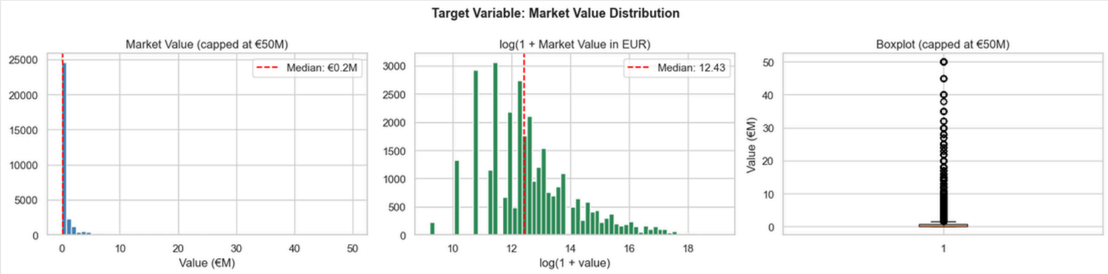
THE PROBLEM

WHY IS THIS HARD?

- Transfer fees exceed **€200M** for top players
- Valuations are **subjective** – clubs, agents, media shape prices
- A striker's value depends on **different signals** than a goalkeeper's

OUR GOAL

- Predict *market_value_in_eur* from **publicly available** performance statistics – **separately per position group**



Market value distribution – heavily right-skewed; log-transform applied

DATA & PIPELINE

7 TRANSFERMARKT CSVS

Source	Size
Players	34,291
Appearances	1.7M rows
Valuations	449K rows
Clubs / Transfers	~85K

After cleaning: 30,710 labelled players

4 positon groups:

GK - DEF - MID- ATT

10-STEP PIPELINE

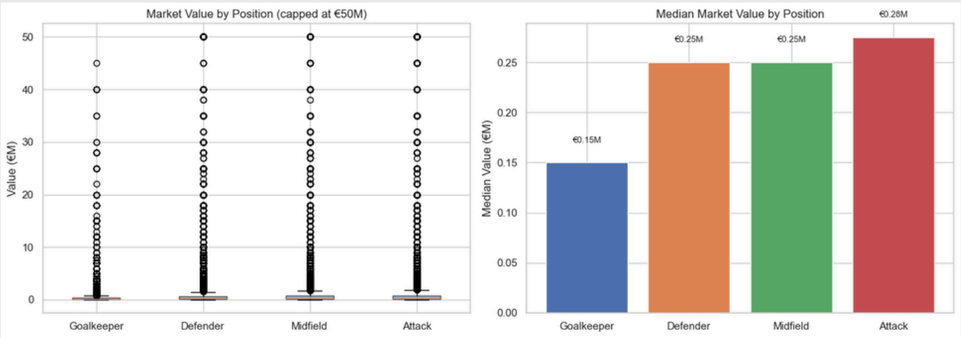
```
Raw CSVs
  ↓ 01 Sanity checks
  ↓ 02 EDA
  ↓ 03 Clean & merge
  ↓ 04 Feature engineering
  ↓ 05 Preprocessing
  ↓ 06 Baseline models
  ↓ 07 Advanced models
  ↓ 08 SHAP & evaluation
  ↓ 09 Undervalued players
  ↓ 10 Final report
```

FEATURE ENGINEERING

SHARED BASE FEATURES (ALL POSITIONS)

- age + age² – non-linear career arc
- log_minutes, log_appearances – experience
- Rate stats: goals, assists, contributions per 90
- log_highest_mv – career peak reputation
- league_mean_value – target-encoded league prestige
- transfer_count, log_total_fees

WHY SEPARATE MODELS?



FEATURE ENGINEERING

POSITION-SPECIFIC ADDITIONS

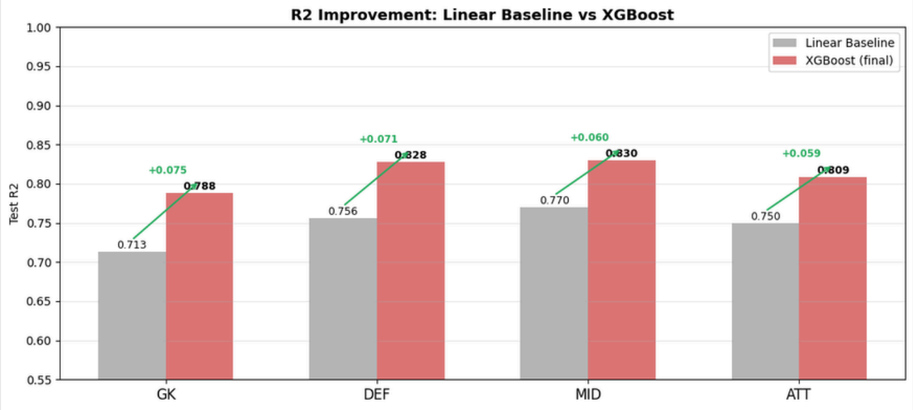
- **GK** - save percentage proxy, clean sheet rate, offensive contribution
- **DEF** - defensive solidity, discipline score, defensive attack rate
- **MID** - creative output per 90, attack-defense ratio
- **ATT** - goals per appearance, minutes per goal, assist-to-goal ratio

MODELS & RESULTS

WHAT WE TRAINED

1. **Linear regression** (plain)
2. **Ridge** (L2 regularisation)
3. **Lasso** (L1 regularisation)
4. **Random Forest** (RandomizedSearchCV)
5. **XGBoost** (hist method, tuned)
6. **MLP** (Adam, early stopping, 3 layers)

No data leakage: split → encode → scale



XGBOOST GAINS +5–8 R^2 POINTS OVER RIDGE BASELINES

TEST-SET R^2 BY POSITION

Position	Ridge (baseline)	Random Forest	MLP	XGBoost (best)
GK	0.713	0.781	0.738	**0.788**
DEF	0.756	0.825	0.815	**0.828**
MID	0.770	0.825	0.819	**0.830**
ATT	0.750	0.802	0.802	**0.809**

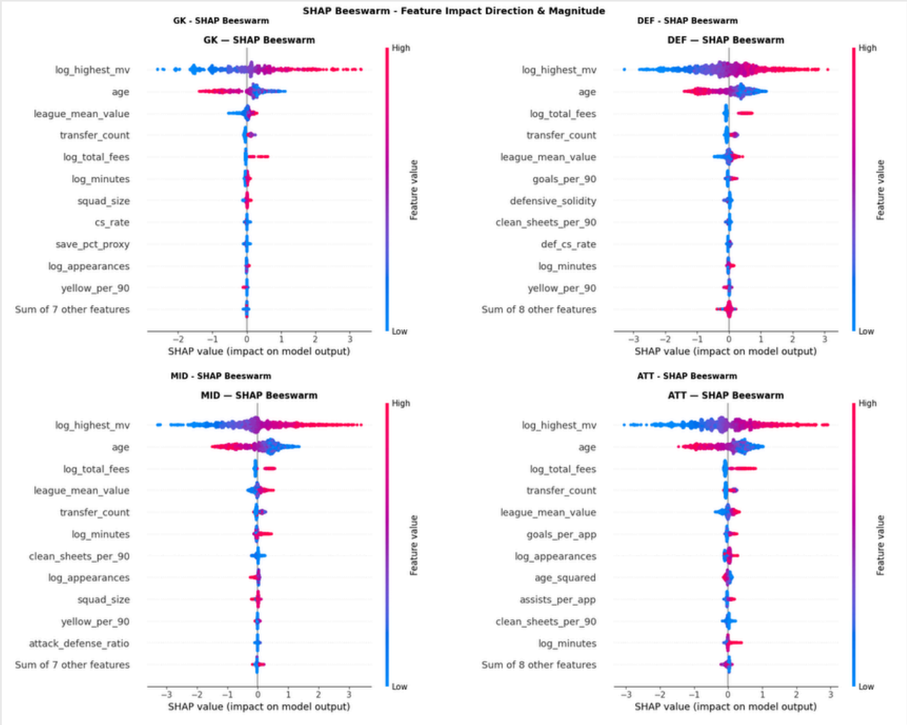
SHAP: WHAT DRIVES VALUE?

CROSS-POSITION FINDINGS

1. **league_mean_value –** top-3 in every position: **where** you play matters as much as **how**
2. **- log_highest_mv –** market anchors to career reputation
3. **- age² –** value peaks mid-20s, accelerates decline post-30

POSITION-SPECIFIC SIGNALS

- **ATT**: goals_per_90 becomes the strongest driver
- **GK**: performance stats matter less; reputation & league dominate
- **MID**: creative_per_90 and contributions_per_90 provide lift
- **DEF**: defensive_solidity and transfer_count add beyond base



APPLICATION: DREAM TEAM

SCORING ALL 33,500+ PLAYERS

- Using serialised preprocessors – **no leakage**
- **Undervaluation ratio** = predicted ÷ actual
- Minimum 10 appearances filter applied

4-3-3 SQUAD – COST VS. PREDICTED VALUE

Metric	Value
Total actual cost	€5.71M
Total predicted value	€60.24M
Return ratio	10.5×

TOP PICKS:
· MAXI GÓMEZ (ATT, ES1, €16.15M GAP)
· CONRAD HARDER (ATT, L1, €15.48M GAP)
· JORIS GNAGNON (DEF, ES1, 24.6×)

LIMITATIONS & FUTURE WORK

Area	Limitation	Suggested Fix
Stars	Q5 (>€20M) has highest RMSE — brand/hype value not captured	Log-linear correction for top percentile
Young players	U22 predictions least reliable — future potential is non-statistical	Add scouting metadata, youth academy flag
Time	No recency — a player on form right now looks same as 3 years ago	Rolling averages, `valuation_year` feature
Injuries	No injury history in dataset	Join injury records from external source
Positions	GK/DEF/MID/ATT too coarse	CB vs LB vs RB, CM vs CAM vs CDM
Uncertainty	Point estimates only	Quantile regression or conformal prediction
Coverage	14 leagues only; players in other leagues fall back to global mean	Expand to more competitions

CONCLUSIONS

WHAT WE ACHIEVED

- R^2 **0.79–0.83** ACROSS ALL POSITIONS
- XGBOOST OUTPERFORMS RF, MLP, AND LINEAR BASELINES
- SHAP PROVIDES ACTIONABLE INTERPRETATION
- FULL REPRODUCIBLE PIPELINE (10 NOTEBOOKS)

KEY INSIGHTS

- LEAGUE PRESTIGE = STRONGEST CROSS-POSITION SIGNAL
- CAREER PEAK VALUE ANCHORS MARKET ESTIMATES
- POSITION-SPECIFIC FEATURES ADD MEANINGFUL LIFT
- HIGH-VALUE PLAYERS SYSTEMATICALLY UNDERESTIMATED (COMPRESSION EFFECT)

FUTURE WORK

- FINER POSITION GRANULARITY (CB/LB/RB, CM/CAM/CDM)
- TEMPORAL MODELLING (LSTM/TRANSFORMER) FOR CAREER TRAJECTORY
- QUANTILE REGRESSION FOR UNCERTAINTY ESTIMATES
- EXPAND BEYOND 14 LEAGUES; JOIN INJURY RECORDS

THANK YOU

- **REPOSITORY:** FOOTBALL-PLAYER-VALUE-PREDICTOR
- **PIPELINE:** 10 NOTEBOOKS · 7 RAW CSVS → DREAM TEAM
- **BEST MODEL:** XGBOOST · R^2 UP TO **0.830** (MID)